

Day – 06

Date: 23 September 2026

Topic: Embeddings & Vector Databases

What I learned:

Embeddings, text-to-vector representation, semantic similarity, vector databases, similarity search, metadata, and the role of embeddings in AI applications.

What I implemented:

Created a small collection of technical documents, converted the documents into embeddings using Sentence Transformers, stored the embeddings and metadata in ChromaDB, and implemented semantic similarity search.

What worked:

The system successfully retrieved semantically relevant documents for queries that did not always use the exact wording contained in the stored documents.

What failed:

The system could still return an unrelated or only partially relevant document when the requested information was not available in the database.

How I solved it:

I investigated the returned similarity-search results and tested different queries. I identified that the vector database searches for the closest available vectors rather than determining whether the database actually contains the correct answer. I documented this as a limitation of the current implementation.

What I would improve:

I would use a larger document collection, experiment with different embedding models, add similarity thresholds and metadata filtering, and investigate reranking techniques.

Screenshots:

```
(venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-TrainingWeek-02\Day-06\practical-task> python embeddings.py
modules.json: 100% ██████████ 349/349 [00:00:00.00, 1.60MB/s]
E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-TrainingWeek-02\Day-06\practical-task\.venv\Lib\site-packages\huggingface_hub\file_download.py:158: UserWarning: 'huggingface_hub' cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\Wlshusha\.cache\huggingfacehub\models--sentence-transformers--all-MiniLM-L6-v2. Caching files will still work but in a degraded version that might require more space on you disk. This warning can be disabled by setting the "HF_HUB_DISABLE_SYMLINKS_WARNING" environment variable. For more details, see https://huggingface.co/docs/huggingfac_e_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: h_ttps://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config_sentence_transformers.json: 100% ██████████ 116/116 [00:00:00.00, 759KB/s]
README.md: 100% ██████████ 10.5k/10.5k [00:00:00.00, 42.5MB/s]
sentence_bert_config.json: 100% ██████████ 53.0/53.0 [00:00:00.00, 283kB/s]
config.json: 100% ██████████ 612/612 [00:00:00.00, 2.46MB/s]
E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-TrainingWeek-02\Day-06\practical-task\.venv\Lib\site-packages\huggingface_hub\file_download.py:158: UserWarning: 'huggingface_hub' cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\Wlshusha\.cache\huggingfacehub\files. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting
```

```

[embedding:
[0.07368828352666022, -0.046505156904559, 0.0269661303609696, 0.03296903148294959, 0.003769845701754093, -0.05885068699717522, 0.017250576987862587, -0.103424467146396
64, -0.025025210888251, -0.031424242902696133, -0.028535978868603706, -0.0301846319938676157, 0.024436043575406075, -0.025347519665956497, -0.053998682727989465, 0.02339
01385217980954, -0.00146760674098162588, 0.021735563271814056, -0.0636065133141632, -0.0968275965976702, 0.0276588574085187, -0.04159459842416055, -0.0512493247851372,
-0.0409765352889924, 0.039416175353645676, 0.0979837236897659, 0.0546133297248001, 0.04173500447723754, 0.06163904041110565, -0.01040752604637674, 0.0574277383980831,
-0.016402488574385643, -0.027804521643791008, 0.013311686596244889, -0.05512169376015663, -0.012881089701552753, -0.0764672681689264, 0.06407493289286438, -0.09944255
129992962, 0.049173954847277115, 0.036611407995224, -0.022983064875086676, 0.004561426118616243, 0.04047737643122673, 0.01794604724645615, 0.02684380422345352, -0.04585
619327132932, -0.0041966611644745, -0.00957902727589397, 0.05864161839835236, -0.0943576589758714, -0.05436392465460494, -0.03909613978518112, -0.018232336717572905,
-0.00883180189586324, 0.050192150831222524, -0.11780864380525406, 0.017293348908424737, 0.0891016687551804, -0.02349862412815860, -0.009606215027689444, -0.102714284550
90487, -0.108860765374661972, -0.0292686615780895, 0.04062587393510402, -0.045136613805794, -0.01022556788986488, 0.0650416690487681, 0.006360464762374629, 0.07965
20156383514, -0.0797914386852805, 0.14385037260055542, -0.010509007297432423, 0.019427904266152302, 0.06153087748755409, 0.039842074556709544, -0.0229538093606589,
0.018873710650895067, 0.0341552532093239, -0.0345178871011734, 0.04171336442232132, 0.0087184598669640875, 0.007435156032434847, 0.04739058274030685, -0.1842309479253086,
-0.058673969786581116, 0.02844960428774357, 0.006195623472629471, 0.0086660077953861, 0.007295916322618732, -0.028607454150911545, -0.06176557714308212, -0.02689446
62392807, -0.07366252690553665, -0.0725104088087373, 0.0889564320343018, -0.0305075533880684, -0.1312787244348145, 0.06844261288628307, 0.0171813945939785, -0.09491
7269516556, 0.047336685246026871, -0.018947949747769165, 0.03829254582524996, -0.0206504312175631, -0.0405365985139847, 0.01413913972398175, 0.0006436987674575448, -0
113973125319154, -0.06632298976182938, -0.02302806353569, 0.063840718746183, -0.050165453431619985, -0.0166088540549578607, -0.0172385364770893, -0.04160065680742
264, -0.03583567572831075, -0.02656658925110602, -0.01247912947367382, 0.11652908171294545, 0.0315669808046032, 0.025924690067768097, -0.01586263371579349, 0.003581
434289932, -0.0140662132537365, -0.0403501851347008, -0.1008471846580554, -2.3036869692384785e-33, 0.04158493893269192, -0.0528433146212578, 0.014848889439363671,
0.01410552962502343, -0.0312088854168653488, -0.00776147624454498, -0.0643564954075813, -0.032471686661638794, 0.081712378919426, 0.062725521417350679, -0.0403635678356

```

[illegible]

```
Query:
How can computers learn from data?

Most Similar Documents:

Result 1
Document: Machine learning allows computers to learn patterns from data and make predictions.
Distance: 0.6618492603302002

Result 2
Document: Vector databases are designed to store and search high-dimensional vectors efficiently.
Distance: 1.349212884902954

Result 3
Document: Python is a popular programming language used for artificial intelligence and data science.
Distance: 1.4717158079147339
(.venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-Training\Week-02\Day-06\practical-task> python search.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%| 103/103 [00:00<00:00, 23
```

```
(.venv) PS E:\VisionRoot-AI engineer Internship\Github\AI-Engineer-Training\AI-Engineer-Training\Week-02\Day-06\practical-task> python app.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|██████████████████████████████████████████████████████████████████████████████| 103/103 [00:00<00:00, 3184.62it/s]
=====
SEMANTIC SEARCH SYSTEM
=====

Enter your search query (or type 'exit'): How does artificial intelligence learn?

Most Relevant Documents:

Result 1
Document: Machine learning allows computers to learn patterns from data and make predictions.
Topic: Machine Learning
Category: AI
Distance: 0.8219241499900818
```